

International Institute of Information Technology, Hyderabad
MA4.101-Real Analysis (Monsoon-2026)
Assignment 1

Question 1 [15 Marks]. Define $\text{pred} : \mathbb{N} \rightarrow \mathbb{N}$ recursively by

$$\text{pred}(0) := 0, \quad \text{pred}(n++) := n,$$

and then define $m \ominus n$ (“ m minus n ”) by recursion on n :

$$m \ominus 0 := m, \quad m \ominus (n++) := \text{pred}(m \ominus n).$$

Prove the following.

- (a) [1 Mark] For all $n \in \mathbb{N}$, $0 \ominus n = 0$.
- (b) [1 Mark] For all $m, n \in \mathbb{N}$, $m++ \ominus n++ = m \ominus n$.
- (c) [2 Marks] $(m \ominus n) \ominus k = m \ominus (n + k)$ for all $m, n, k \in \mathbb{N}$.
- (d) [1 Mark] $(m + n) \ominus n = m$ for all $m, n \in \mathbb{N}$.
- (e) [2 Marks] $m \ominus n = 0$ if and only if $m \leq n$.
- (f) [2 Marks] $(m \ominus n) + n = m$ if and only if $n \leq m$.
- (g) [2 Marks] Define $\max(m, n) := m$ if $n \leq m$, and $\max(m, n) := n$ otherwise. Then $\max(m, n) = m + (n \ominus m)$, and deduce that $m + (n \ominus m) = n + (m \ominus n)$.
- (h) [2 Marks] $k \times (m \ominus n) = (k \times m) \ominus (k \times n)$.
- (i) [2 Marks] Suppose $k \leq m$ and $k \leq n$. Prove that $m \ominus k = n \ominus k$ implies $m = n$. Show that neither hypothesis can be dropped.

Solution. Throughout, $m \leq n$ means that $n = m + k$ for some $k \in \mathbb{N}$, and we use the trichotomy of order: for any m, n at least one of $m \leq n$ and $n \leq m$ holds, and both hold exactly when $m = n$.

(a) Induct on n . For $n = 0$, $0 \ominus 0 = 0$ by the first clause of the definition of $\ominus$. If $0 \ominus n = 0$ then $0 \ominus n++ = \text{pred}(0 \ominus n) = \text{pred}(0) = 0$, using the second clause of $\ominus$ and then the first clause of pred .

(b) Induct on n , with m fixed. For $n = 0$,

$$m++ \ominus 0++ = \text{pred}(m++ \ominus 0) = \text{pred}(m++) = m = m \ominus 0.$$

If $m++ \ominus n++ = m \ominus n$, then

$$m++ \ominus n++++ = \text{pred}(m++ \ominus n++) = \text{pred}(m \ominus n) = m \ominus n++.$$

(c) Induct on k , with m and n fixed. For $k = 0$ both sides equal $m \ominus n$. Assuming the identity for k ,

$$(m \ominus n) \ominus k++ = \text{pred}((m \ominus n) \ominus k) = \text{pred}(m \ominus (n+k)) = m \ominus (n+k)++ = m \ominus (n+k++),$$

using the definition of $\ominus$ twice and $(n+k)++ = n+k++$.

(d) Induct on n , with m fixed. For $n = 0$, $(m+0) \ominus 0 = m$. Assuming the identity for n , part (b) gives

$$(m+n++) \ominus n++ = (m+n)++ \ominus n++ = (m+n) \ominus n = m.$$

(e) First, $m \ominus m = 0$: apply part (d) with m replaced by 0, giving $(0+m) \ominus m = 0$.
 $(\Leftarrow)$ Suppose $m \leq n$, say $n = m+k$. By parts (a)–(c), we have

$$m \ominus n = m \ominus (m+k) = (m \ominus m) \ominus k = 0 \ominus k = 0.$$

$(\Rightarrow)$ We prove the contrapositive. If $m \not\leq n$ then, by trichotomy, $n \leq m$ and $n \neq m$, so $m = n+k$ with $k \neq 0$. By part (d),

$$m \ominus n = (k+n) \ominus n = k \neq 0. \quad \square$$

(f) $(\Leftarrow)$ If $n \leq m$, write $m = n+j$. By part (d), $m \ominus n = (j+n) \ominus n = j$, so $(m \ominus n) + n = j + n = m$.

$(\Rightarrow)$ If $(m \ominus n) + n = m$, then $m = n + (m \ominus n)$, so m is n plus something, which is exactly the definition of $n \leq m$.

(g) The definition of $\max$ splits on whether $n \leq m$, so we take those two cases.

Case 1: $n \leq m$. Then $\max(m, n) = m$ by the first clause of the definition. By part (e), $n \ominus m = 0$, so

$$m + (n \ominus m) = m + 0 = m = \max(m, n).$$

Case 2: $n \not\leq m$. Then $\max(m, n) = n$ by the second clause. By trichotomy $m \leq n$, say $n = m + k$, and part (d) gives $n \ominus m = (k + m) \ominus m = k$. Hence

$$m + (n \ominus m) = m + k = n = \max(m, n).$$

For the deduction we first check that $\max$ is symmetric, which is not immediate from the definition, since the definition tests only whether $n \leq m$. By trichotomy there are three cases.

- If $n \leq m$ and $n \neq m$, then $\max(m, n) = m$; also $m \not\leq n$, so the second clause gives $\max(n, m) = m$.
- If $m \leq n$ and $m \neq n$, the same argument with the roles exchanged gives $\max(m, n) = n = \max(n, m)$.
- If $m = n$, then $n \leq m$ and $m \leq n$ both hold, so $\max(m, n) = m = n = \max(n, m)$.

In every case $\max(m, n) = \max(n, m)$. Applying the identity just proved twice, once as stated and once with m and n exchanged,

$$m + (n \ominus m) = \max(m, n) = \max(n, m) = n + (m \ominus n).$$

(h) By trichotomy at least one of $n \leq m$ and $m \leq n$ holds, so it suffices to treat those two cases.

If $n \leq m$, write $m = n + j$, so that $m \ominus n = j$ by part (d). Then $km = kn + kj$ by distributivity, so part (d) applied again gives

$$km \ominus kn = (kj + kn) \ominus kn = kj = k(m \ominus n).$$

If $m \leq n$, then $m \ominus n = 0$ by part (e), so the right side is $k \times 0 = 0$. Also $km \leq kn$, since multiplication preserves order, so the left side is 0 by part (e) as well.

(i) By part (f), $k \leq m$ gives $(m \ominus k) + k = m$, and $k \leq n$ gives $(n \ominus k) + k = n$. Hence

$$m = (m \ominus k) + k = (n \ominus k) + k = n.$$

Neither hypothesis can be dropped on its own. The statement is symmetric under exchanging m and n , so a single example settles both directions. Take

$$k := 0++, \quad m := 0++, \quad n := 0.$$

Here $k \leq m$ holds, since $m = k + 0$, whereas $k \leq n$ fails, since $0++ \neq 0 + j$ for any j would force $0++ = j$ and then $0 \leq 0++$ with $0++ \leq 0$, giving $0 = 0++$, contradicting the axiom that 0 is not a successor. Now $m \ominus k = 0++ \ominus 0++ = 0$ by the first line of part (e), and $n \ominus k = 0 \ominus 0++ = 0$ by part (a). So $m \ominus k = n \ominus k$ while $m \neq n$. Exchanging the roles of m and n , that is taking $k := 0++$, $m := 0$, $n := 0++$, gives an example in which $k \leq n$ holds and $k \leq m$ fails, with the same computation. Hence neither hypothesis alone can be omitted.

Question 2 [10 Marks]. We wish to define $h : \mathbb{N} \rightarrow \mathbb{N}$, the “lower half” of n , by

$$h(0) := 0, \quad h(0++) := 0, \quad h(n+++++) := h(n)++. \quad (1)$$

- (a) **[2 Marks]** Prove that every $n \in \mathbb{N}$ is of exactly one of the forms 0 , $0++$, or $m+++++$, and that in the last case m is uniquely determined by n . Explain why this is needed before Eq. (1) can be read as a definition at all.
- (b) **[2 Marks]** Explain why Eq. (1) is nevertheless not an instance of the recursion theorem.
- (c) **[4 Marks]** Prove that a function $h : \mathbb{N} \rightarrow \mathbb{N}$ satisfying Eq. (1) exists. State clearly which set you recurse into, what the step rule is, and how h itself is recovered from the function the recursion theorem gives you.
- (d) **[2 Marks]** Prove that such an h is unique.

Solution.

(a) We use Tao’s Axiom 2.3 (0 is not the successor of any natural number), Axiom 2.4 (the successor is injective: $x++ = y++$ implies $x = y$), and the principle of mathematical induction (Axiom 2.5).

Every n has one of the three forms. Induct on n . For $n = 0$ the first form applies. Suppose n has one of the three forms and consider $n++$. If $n = 0$ then $n++ = 0++$, the second form. If $n = 0++$ then $n++ = 0+++++$, the third form with $m = 0$. If $n = m+++++$ then $n++ = (m++)+++++$, the third form with $m++$ in place of m . So the claim passes from n to $n++$, and by induction it holds for every n .

The three forms are mutually exclusive. If $n = 0$ then n is not a successor (Axiom 2.3), so it is neither $0++$ nor of the form $m+++++$. If $0++ = m+++++$ then Axiom

2.4 gives $0 = m++$, so 0 would be a successor, contradicting Axiom 2.3. Hence no n has two of the three forms.

Uniqueness of m . If $m++++ = m'++++$ then two applications of Axiom 2.4 give $m = m'$.

Why this is needed. For Eq. (1) to prescribe a function at all, every $n \in \mathbb{N}$ must be assigned a value, and no n may be assigned two. Exhaustiveness supplies the first: every n falls under at least one of the three clauses. Mutual exclusivity supplies half of the second: no n falls under two clauses at once, which would risk $h(n)$ being prescribed as both 0 and something else. Uniqueness of m supplies the other half: without it, an n of the form $m++++$ might also equal $m'++++$ for some $m' \neq m$, and the third clause would prescribe both $h(m)++$ and $h(m')++$ at that single point. Note that splitting on “0, or a successor of 0, or a double successor” is not itself one of the axioms; it is the statement proved here.

(b) The recursive definition of Tao takes as input a natural number c and, for each $n \in \mathbb{N}$, a function $f_n : \mathbb{N} \rightarrow \mathbb{N}$, and returns the unique sequence $(a_n)_{n \in \mathbb{N}}$ of natural numbers with

$$a_0 = c, \quad a_{n++} = f_n(a_n) \quad \text{for all } n.$$

Two features of that shape are absent from Eq. (1). First, the recursive definition takes a single initial value, whereas Eq. (1) prescribes two, at 0 and at $0++$. Second, and decisively, the proposition determines the value at each successor $n++$ from the value at the immediately preceding number n , whereas the third clause of Eq. (1) relates $h(n++++)$ to $h(n)$, two steps back, and says nothing whatever about $h(n++)$ in terms of $h(n)$.

To present Eq. (1) as an instance we would have to exhibit a family $(f_n)_{n \in \mathbb{N}}$ with $h(n++) = f_n(h(n))$ for every n , and Eq. (1) supplies none. Nor can such a family be read off from Eq. (1) directly: whether $h(n++)$ equals $h(n)$ or $h(n)++$ depends on n (informally, on its parity), and nothing in Eq. (1) says which case occurs for a given n . So the existence of h does not follow by merely citing Proposition 2.1.16; a suitable family (f_n) has to be constructed first, which is done in (c).

(c) *The set recursed into.* The recursive definition only ever produces sequences of natural numbers, so the set recursed into is $\mathbb{N}$ in every application; there is no option of recursing into a larger set such as $\mathbb{N} \times \mathbb{N}$. Instead we exploit the one extra freedom the proposition does allow: the step rule f_n may depend on the index n . We apply the proposition twice, first to build a parity indicator, and then to build h .

First application: a parity indicator. Recall $1 := 0++$. Define $\varphi : \mathbb{N} \rightarrow \mathbb{N}$ by

$$\varphi(y) := 1 \quad \text{if } y = 0, \quad \varphi(y) := 0 \quad \text{if } y \neq 0.$$

This assigns exactly one value to each $y \in \mathbb{N}$, since exactly one of $y = 0$ and $y \neq 0$ holds. Apply recursive definition with $c := 0$ and with $f_n := \varphi$ for every n . This gives

a unique sequence $(p_n)_{n \in \mathbb{N}}$ with

$$p_0 = 0, \quad p_{n++} = \varphi(p_n) \quad \text{for all } n. \quad (2)$$

This is a genuine one-step recursion: p_{n++} depends only on p_n . Note the two properties.

(P1) For every n , either $p_n = 0$ or $p_n = 1$. Indeed, $p_0 = 0$, and for any n the value $p_{n++} = \varphi(p_n)$ is 0 or 1 by the definition of φ ; so the claim follows by induction.

(P2) If $p_n = 0$ then $p_{n++} = \varphi(0) = 1$. If $p_n = 1$ then, since $1 = 0++ \neq 0$ by Axiom 2.3, $p_{n++} = \varphi(1) = 0$.

(Informally, p_n is 0 when n is even and 1 when n is odd, but we never need this interpretation; only (P1) and (P2) are used below.)

Second application: the step rule and h . For each $n \in \mathbb{N}$ define $f_n : \mathbb{N} \rightarrow \mathbb{N}$ by

$$f_n(y) := y \quad \text{if } p_n = 0, \quad f_n(y) := y++ \quad \text{if } p_n \neq 0.$$

For each fixed n exactly one of the two cases applies, so each f_n assigns exactly one value to each y , and is a function from $\mathbb{N}$ to $\mathbb{N}$. Apply recursive definition with $c := 0$ and this family (f_n) . This gives a unique sequence $(a_n)_{n \in \mathbb{N}}$ with

$$a_0 = 0, \quad a_{n++} = f_n(a_n) \quad \text{for all } n. \quad (3)$$

Recovering h . Set $h(n) := a_n$ for every $n \in \mathbb{N}$. Here h is the sequence produced by the recursive definition itself. The extra piece of memory needed to reach two steps back is carried not by the value a_n but by the index n : through p_n , the rule f_n knows whether this is a step at which h stays the same or increases by one.

Verification of the three clauses. Using Eq. (3):

- $h(0) = a_0 = 0$.
- $h(0++) = a_{0++} = f_0(a_0) = a_0 = 0$, since $p_0 = 0$.
- Let $n \in \mathbb{N}$. Applying Eq. (3) twice,

$$h(n++++) = a_{n++++} = f_{n++}(a_{n++}) = f_{n++}(f_n(h(n))).$$

By (P1) there are two cases. If $p_n = 0$, then $f_n(h(n)) = h(n)$, and by (P2) $p_{n++} = 1 \neq 0$, so $f_{n++}(h(n)) = h(n)++$. If $p_n = 1$, then $p_n \neq 0$, so $f_n(h(n)) = h(n)++$, and by (P2) $p_{n++} = 0$, so $f_{n++}(h(n)++) = h(n)++$. In either case $h(n++++) = h(n)++$.

So h satisfies Eq. (1), and existence is proved.

(d) Suppose h and h' both satisfy Eq. (1). A plain induction on the statement $h(n) = h'(n)$ does not close, because the third clause relates the value at $n++++$ to the value at n rather than at $n++$. We therefore strengthen the statement being inducted on. Let $Q(n)$ be

$$h(n) = h'(n) \text{ and } h(n++) = h'(n++).$$

Base case. $h(0) = 0 = h'(0)$ and $h(0++) = 0 = h'(0++)$, by the first two clauses of Eq. (1) applied to each of h and h' . So $Q(0)$ holds.

Inductive step. Assume $Q(n)$. The first half of $Q(n++)$ is the assertion $h(n++) = h'(n++)$, which is precisely the second half of $Q(n)$. For the second half of $Q(n++)$, the third clause of Eq. (1) and the first half of $Q(n)$ give

$$h(n++++) = h(n)++ = h'(n)++ = h'(n++++).$$

So $Q(n++)$ holds.

By induction $Q(n)$ holds for every $n \in \mathbb{N}$, and in particular $h(n) = h'(n)$ for every n . So h and h' assign the same natural number to every n , that is, $h = h'$. $\square$

Question 3 [8 Marks]. Let $n = 2^k$ for some $k \in \mathbb{N}$. Prove that for all $a_1, \dots, a_n \in \mathbb{N}$,

$$(a_1 + a_2 + \dots + a_n)^n \geq n^n a_1 a_2 \dots a_n.$$

Solution. We use freely that finite sums and products of natural numbers obey the usual algebra, and that multiplication preserves order: if $x \geq y$ then $xz \geq yz$, and consequently $x \geq y$ implies $x^j \geq y^j$ for every $j \in \mathbb{N}$, by induction on j .

A two-variable lemma. For all $S, T \in \mathbb{N}$,

$$S^2 + T^2 \geq 2ST.$$

The statement is symmetric in S and T , so by trichotomy we may assume $T \leq S$ and write $S = T + u$ with $u \in \mathbb{N}$. Then

$$S^2 + T^2 = (T + u)^2 + T^2 = 2T^2 + 2Tu + u^2, \quad 2ST = 2(T + u)T = 2T^2 + 2Tu,$$

so the left side exceeds the right by exactly $u^2 \geq 0$. Adding $2ST$ to both sides of the lemma gives

$$(S + T)^2 = S^2 + 2ST + T^2 \geq 4ST. \tag{4}$$

Note that the customary proof, from $(S - T)^2 \geq 0$, is unavailable here, since $S - T$ need not lie in $\mathbb{N}$; writing the larger as the smaller plus something is the substitute.

The induction. Induct on k . For $k = 0$ we have $n = 2^0 = 1$ and the claim reads $a_1 \geq a_1$, which holds.

Assume the claim for $n = 2^k$, and let $a_1, \dots, a_{2n} \in \mathbb{N}$. Put

$$S := a_1 + \dots + a_n, \quad T := a_{n+1} + \dots + a_{2n}.$$

Raising Eq. (4) to the power n , which preserves the inequality,

$$(S + T)^{2n} = ((S + T)^2)^n \geq (4ST)^n = 4^n S^n T^n.$$

The induction hypothesis applies to each half, giving $S^n \geq n^n \prod_{i=1}^n a_i$ and $T^n \geq n^n \prod_{i=n+1}^{2n} a_i$. Multiplying these two inequalities together, which is legitimate because multiplication preserves order in each argument separately, and then multiplying by 4^n ,

$$4^n S^n T^n \geq 4^n \left(n^n \prod_{i=1}^n a_i \right) \left(n^n \prod_{i=n+1}^{2n} a_i \right) = 4^n n^n n^n \prod_{i=1}^{2n} a_i.$$

Finally $4^n n^n n^n = (4n^2)^n = ((2n)^2)^n = (2n)^{2n}$, so chaining the two displays,

$$\left(\sum_{i=1}^{2n} a_i \right)^{2n} \geq (2n)^{2n} \prod_{i=1}^{2n} a_i,$$

which is the claim for $2n = 2^{k++}$. $\square$

Remark on the hypothesis. This is the arithmetic–geometric mean inequality with the denominators cleared, so that it becomes a statement about natural numbers. Restricting to powers of two is what keeps the induction purely forward: each step doubles the number of variables. It is a convenience rather than a necessity. The classical route to arbitrary n descends from 2^k by replacing a variable with the average of the others, and averages do not exist in $\mathbb{N}$; but the cleared-denominator form admits a subtraction-free descent. Suppose the result holds for n variables and let $a_1, \dots, a_{n \ominus 1} \in \mathbb{N}$ with $s := a_1 + \dots + a_{n \ominus 1}$. Applying the n -variable case to the n numbers

$$(n \ominus 1)a_1, \dots, (n \ominus 1)a_{n \ominus 1}, s,$$

whose sum is $(n \ominus 1)s + s = ns$, yields $n^n s^n \geq n^n (n \ominus 1)^{n \ominus 1} s \prod_i a_i$. Cancelling n^n , and then cancelling s when $s \neq 0$ (if $s = 0$ every a_i is 0 and the claim is $0 \geq 0$), gives exactly the statement for $n \ominus 1$ variables. Since the powers of two are unbounded, every n is reached. A solution that carries this out deserves full credit and more.

Question 4 [8 Marks]. Suppose $a_1 \leq a_2 \leq \dots \leq a_n$ and $b_1 \leq b_2 \leq \dots \leq b_n$ are

natural numbers. Prove that

$$n \sum_{i=1}^n a_i b_i \geq \left(\sum_{i=1}^n a_i \right) \left(\sum_{i=1}^n b_i \right).$$

Solution. *Pairwise lemma.* For all indices i, j ,

$$a_i b_i + a_j b_j \geq a_i b_j + a_j b_i.$$

The claim is symmetric in i and j , so assume $j \leq i$. Both sequences are increasing, so $a_j \leq a_i$ and $b_j \leq b_i$; write $a_i = a_j + u$ and $b_i = b_j + v$ with $u, v \in \mathbb{N}$. Then

$$a_i b_i + a_j b_j = 2a_j b_j + a_j v + u b_j + uv, \quad a_i b_j + a_j b_i = 2a_j b_j + u b_j + a_j v,$$

so the first exceeds the second by exactly $uv \geq 0$. This is the only point in the argument at which the hypothesis is used, and what it supplies is that $a_j \leq a_i$ and $b_j \leq b_i$ hold *simultaneously*.

Summation. We use the fact that an inequality may be summed: if $x_{ij} \geq y_{ij}$ for all i, j then $\sum_{i,j} x_{ij} \geq \sum_{i,j} y_{ij}$. This follows from the two-term case $x \geq y$ and $x' \geq y'$ imply $x + x' \geq y + y'$, by induction on the number of terms. It is a fact of a different kind from the identities used below to rearrange sums, and is worth stating separately.

Summing the lemma over all i and all j , the left side becomes

$$\sum_i \sum_j (a_i b_i + a_j b_j) = \sum_i \left(\sum_j a_i b_i \right) + \sum_i \left(\sum_j a_j b_j \right) = n \sum_i a_i b_i + n \sum_j a_j b_j = 2n \sum_i a_i b_i,$$

where the inner sum in the first term has n equal summands and the inner sum in the second term does not depend on i . The right side becomes

$$\sum_i \sum_j (a_i b_j + a_j b_i) = \left(\sum_i a_i \right) \left(\sum_j b_j \right) + \left(\sum_j a_j \right) \left(\sum_i b_i \right) = 2 \left(\sum_i a_i \right) \left(\sum_i b_i \right),$$

by distributivity. Hence $2n \sum a_i b_i \geq 2 \left(\sum a_i \right) \left(\sum b_i \right)$, and cancelling the factor 2 — legitimate because $2x \geq 2y$ implies $x \geq y$ in $\mathbb{N}$ — gives the result. $\square$

Remark. No induction on n was needed beyond the standard facts about finite sums. Note also that the proof consumes less than is assumed: it never needs either sequence to be sorted, only that the two agree on which of any two indices carries the larger entry. If instead $a_1 \leq \dots \leq a_n$ and $b_1 \geq \dots \geq b_n$, the same pairwise step with the two sides exchanged reverses the inequality.

Question 5 [10 Marks]. Show that there is no function $f : \mathbb{N} \rightarrow \mathbb{N}$ satisfying

$$f(f(n)) = n++ \quad \text{for every } n \in \mathbb{N},$$

but that there is one satisfying $f(f(n)) = n++++$ for every n .

Solution. Suppose such an f exists.

Step 1: $f(n++) = f(n)++$. Apply f to both sides of $f(f(n)) = n++$ to get $f(f(f(n))) = f(n++)$. On the other hand, the defining identity applied at $f(n)$ in place of n reads $f(f(f(n))) = f(n)++$. Comparing the two gives $f(n++) = f(n)++$.

Step 2: $f(n) = f(0) + n$. Induct on n . The case $n = 0$ is $f(0) = f(0) + 0$. If $f(n) = f(0) + n$, then by Step 1,

$$f(n++) = f(n)++ = (f(0) + n)++ = f(0) + n++.$$

Step 3: contradiction. Write $c := f(0)$. By Step 2 applied twice, first at $f(n)$ and then at n ,

$$n++ = f(f(n)) = c + f(n) = c + c + n.$$

Since $n++ = 0++ + n$, cancelling n gives $c + c = 0++$. But if $c = 0$ then $c + c = 0 \neq 0++$, since 0 is not a successor; and if $c \neq 0$ then $0++ \leq c$, so $c + c \geq 0++++ > 0++$. Either way we have a contradiction, so no such f exists.

The second part. Take $f(n) := n++$, which is a function $\mathbb{N} \rightarrow \mathbb{N}$. Then $f(f(n)) = f(n++) = n++++$ for every n , as required. So the obstruction in the first part is genuinely about the number $0++$ and not about composition. $\square$

Question 6 [8 Marks]. Let X be a set. Prove that $\{X, \mathcal{P}(X), \mathcal{P}(\mathcal{P}(X)), \dots\}$ is a set.

Solution. The ellipsis in the statement is not an abbreviation that may be taken for granted: supplying its meaning is the first half of the problem, and a solution that begins by naming the collection and treating it as given has assumed what it was asked to produce. Accordingly we do not name it until Stage 2, where it is produced.

Conventions. Sets are objects (Axiom 1), so an ordered pair $\langle k, Y \rangle$ with $k \in \mathbb{N}$ and Y a set is a legitimate object. We use the following facts about the order on $\mathbb{N}$ freely: $\leq$ is transitive; $k \leq 0$ implies $k = 0$; $n < n++$; and $k \leq n++$ implies $k \leq n$ or $k = n++$.

Why the obvious tools are unavailable. Specification (Axiom 5) carves a subset out of a set already known to exist, and here no set is known in advance to contain all the iterated power sets — producing one is exactly the problem. Replacement (Axiom 6) is the axiom that transports $\mathbb{N}$ along a rule, and it is what we shall use; but it requires a property $P(n, Y)$ that determines Y uniquely from n , so the rule must be made precise first. The recursion theorem does not supply that rule either: it produces a

function into a target set fixed in advance, and no such target set is available here. Stage 1 therefore carries out the recursion by hand.

Stage 1: making the rule precise.

Definition. Let $n \in \mathbb{N}$. A set s is an n -tower if

- (i) every element of s is an ordered pair $\langle k, Y \rangle$ with $k \in \mathbb{N}$, $k \leq n$, and Y a set; That is s may have elements of the form $\langle 1, Y_{11} \rangle, \langle 1, Y_{12} \rangle, \dots, \langle n, Y_{n1} \rangle, \dots$.
- (ii) for every $k \leq n$ there is exactly one set Y with $\langle k, Y \rangle \in s$; we write $s(k)$ for this Y ; That is s may have elements of the form $\langle 1, Y_1 \rangle, \langle 2, Y_2 \rangle, \dots, \langle n, Y_n \rangle$. Then $s(i) = Y_i$.
- (iii) $s(0) = X$;
- (iv) $s(k++) = \mathcal{P}(s(k))$ for every k with $k++ \leq n$.

Clauses (i) and (ii) say precisely that s is the graph of a function on $\{k \in \mathbb{N} : k \leq n\}$; (iii) and (iv) say that this function obeys the intended recursion as far as it goes. Note that (iv) is vacuous when $n = 0$: if $k++ \leq 0$ then $k++ = 0$, contradicting Axiom 2.3 (0 is not a successor).

Lemma 1 (Restriction). *Let $m \leq n$ and let s be an n -tower. Put*

$$s^{\leq m} := \{ p \in s : p = \langle k, Y \rangle \text{ for some } k \leq m \text{ and some set } Y \}.$$

Then $s^{\leq m}$ is an m -tower, and $s^{\leq m}(k) = s(k)$ for every $k \leq m$.

Proof. The collection $s^{\leq m}$ is a set by specification (Axiom 5), applied to the set s and the displayed property. We check (i)–(iv).

(i) is immediate, since every element of $s^{\leq m}$ is an element of s of the required form with $k \leq m$.

(ii) Let $k \leq m$. Then $k \leq n$ by transitivity, so by (ii) for s there is exactly one Y with $\langle k, Y \rangle \in s$, namely $Y = s(k)$. This pair lies in $s^{\leq m}$ by the defining property, and conversely every pair of $s^{\leq m}$ with first coordinate k lies in s . So it is the unique one, and $s^{\leq m}(k) = s(k)$.

(iii) $0 \leq m$, so $s^{\leq m}(0) = s(0) = X$.

(iv) Suppose $k++ \leq m$. Then $k \leq k++ \leq m$, and also $k++ \leq n$ by transitivity, so $s^{\leq m}(k++) = s(k++) = \mathcal{P}(s(k)) = \mathcal{P}(s^{\leq m}(k))$. $\square$

Lemma 2 (Existence and uniqueness of towers). *For every $n \in \mathbb{N}$ there is exactly one n -tower. We denote it s_n .*

Proof. Induction on n (Axiom 2.5).

Base. Put $s_0 := \{\langle 0, X \rangle\}$, a set by the singleton axiom (Axiom 3). Clause (i) holds, as $0 \leq 0$; clause (ii) holds because the only k with $k \leq 0$ is $k = 0$, and X is the only second coordinate occurring; clause (iii) reads $s_0(0) = X$, which is how s_0 was built; and (iv) is vacuous, as noted above. For uniqueness, let t be any 0-tower. By (i), every element of t is a pair $\langle k, Y \rangle$ with $k \leq 0$, hence $k = 0$; by (ii) and (iii) the only such element is $\langle 0, X \rangle$. So t and s_0 have the same elements, and $t = s_0$ by extensionality.

Step. Suppose s_n is the unique n -tower. Put $Z := \mathcal{P}(s_n(n))$, a set by the power set axiom, and

$$s_{n++} := s_n \cup \{\langle n++, Z \rangle\},$$

a set by the singleton axiom and pairwise union. We check that it is an $n++$ -tower.

(i) Elements of s_n are pairs at some $k \leq n \leq n++$, and the new element is a pair at $n++ \leq n++$.

(ii) Let $k \leq n++$. If $k \leq n$, then s_n contains exactly one pair at k , and the new pair is not one of them, since $k \leq n < n++$ gives $k \neq n++$. If $k = n++$, then no element of s_n is a pair at k , for the same reason, so the new pair is the only one. Either way there is exactly one, and moreover $s_{n++}(k) = s_n(k)$ for $k \leq n$, while $s_{n++}(n++) = Z$.

(iii) $s_{n++}(0) = s_n(0) = X$, using $0 \leq n$.

(iv) Let $k++ \leq n++$. If $k++ \leq n$ then $k \leq n$ as well, and the values at k and $k++$ are inherited from s_n , so the clause holds because it holds for s_n . Otherwise $k++ = n++$, whence $k = n$ by injectivity of the successor (Axiom 2.4), and then

$$s_{n++}(k++) = s_{n++}(n++) = Z = \mathcal{P}(s_n(n)) = \mathcal{P}(s_{n++}(n)) = \mathcal{P}(s_{n++}(k)).$$

For uniqueness, let t be any $n++$ -tower. By Lemma 1 with $m := n$, the set $t^{\leq n}$ is an n -tower, so $t^{\leq n} = s_n$ by the induction hypothesis, and $t(k) = s_n(k)$ for every $k \leq n$. By (i), every element of t is a pair at some $k \leq n++$, and each such k satisfies $k \leq n$ or $k = n++$; hence

$$t = t^{\leq n} \cup \{\langle n++, t(n++) \rangle\} = s_n \cup \{\langle n++, t(n++) \rangle\}.$$

Applying (iv) to t with $k := n$ gives $t(n++) = \mathcal{P}(t(n)) = \mathcal{P}(s_n(n)) = Z$. Hence $t = s_{n++}$, closing the induction. $\square$

Definition. For $n \in \mathbb{N}$ put $\mathcal{P}^n(X) := s_n(n)$.

Lemma 3 (The rule is the intended one). $\mathcal{P}^0(X) = X$, and $\mathcal{P}^{n++}(X) = \mathcal{P}(\mathcal{P}^n(X))$ for every $n \in \mathbb{N}$.

Proof. The first identity is $s_0(0) = X$, which is clause (iii) for s_0 . For the second, $s_{n++}^{\leq n}$ is an n -tower by Lemma 1, so $s_{n++}^{\leq n} = s_n$ by Lemma 2, and therefore $s_{n++}(n) = s_n(n) = \mathcal{P}^n(X)$. Since $n++ \leq n++$, clause (iv) for s_{n++} with $k := n$ gives

$$\mathcal{P}^{n++}(X) = s_{n++}(n++) = \mathcal{P}(s_{n++}(n)) = \mathcal{P}(\mathcal{P}^n(X)). \quad \square$$

So $\mathcal{P}^0(X), \mathcal{P}^1(X), \mathcal{P}^2(X), \dots$ are $X, \mathcal{P}(X), \mathcal{P}(\mathcal{P}(X)), \dots$, and the ellipsis of the statement now has a precise meaning: it stands for the objects $\mathcal{P}^n(X)$, one for each $n \in \mathbb{N}$.

Stage 2: applying replacement. Let $P(n, Y)$ be the property

“there is an n -tower s with $\langle n, Y \rangle \in s$ ”.

Fix $n \in \mathbb{N}$. Taking $s := s_n$ and $Y := s_n(n)$ shows that some Y satisfies $P(n, Y)$. Conversely, suppose $P(n, Y)$ holds, witnessed by an n -tower s . By Lemma 2, $s = s_n$; and by clause (ii) for s_n with $k := n$, there is exactly one Y with $\langle n, Y \rangle \in s_n$, namely $s_n(n) = \mathcal{P}^n(X)$. So for each $n \in \mathbb{N}$ there is exactly one Y with $P(n, Y)$, which is what replacement requires.

Replacement (Axiom 6), applied with $A := \mathbb{N}$ (a set by the axiom of infinity) and the property P , now gives that

$$\mathcal{T} := \{ Y : P(n, Y) \text{ for some } n \in \mathbb{N} \} = \{ \mathcal{P}^n(X) : n \in \mathbb{N} \}$$

is a set. By Lemma 3 its elements are exactly $X, \mathcal{P}(X), \mathcal{P}(\mathcal{P}(X)), \dots$, so $\mathcal{T}$ is the collection of the statement, and it is a set. ■

Question 7 [6 Marks]. Let X be a set. Prove that no function $f : X \rightarrow \mathcal{P}(X)$ is surjective.

Solution. Let $f : X \rightarrow \mathcal{P}(X)$ be any function. By the power set axiom $\mathcal{P}(X)$ is a set, so the statement makes sense; note that for each $x \in X$ the value $f(x)$ is an element of $\mathcal{P}(X)$, that is, a subset of X , so the condition $x \notin f(x)$ is meaningful. By specification (axiom 5) applied to the set X and the property $x \notin f(x)$,

$$D := \{ x \in X : x \notin f(x) \}$$

is a set, and $D \subseteq X$, so $D \in \mathcal{P}(X)$.

Suppose, for contradiction, that f were surjective. Since $D \in \mathcal{P}(X)$, there would be some $a \in X$ with $D = f(a)$. Now

$$a \in D \iff a \notin f(a) \iff a \notin D,$$

where the first equivalence is the definition of D , using $a \in X$, and the second substitutes $f(a) = D$. So $a \in D$ holds if and only if it fails, which is impossible. Hence no such a exists, and f is not surjective. □

Remark. Since $x \mapsto \{x\}$ is an injection $X \rightarrow \mathcal{P}(X)$, the result says that $\mathcal{P}(X)$ is strictly larger than X , and iterating it produces unboundedly many sizes of infinity.

The set D is built to disagree with $f(x)$ at x for every x at once, which is why it can be no value of f ; this is Russell's construction, with $f(x)$ in place of x itself.

Question 8 [10 Marks]. If X is a subset of Y , let $\text{inc}_{X \rightarrow Y} : X \rightarrow Y$ be the *inclusion map* from X to Y , defined by

$$\text{inc}_{X \rightarrow Y}(x) := x \quad \text{for all } x \in X.$$

The map $\text{inc}_{X \rightarrow X}$ is called the *identity map* on X . Prove the following.

- (a) [2 Marks] If $X \subseteq Y \subseteq Z$, then $\text{inc}_{Y \rightarrow Z} \circ \text{inc}_{X \rightarrow Y} = \text{inc}_{X \rightarrow Z}$.
- (b) [2 Marks] If $f : A \rightarrow B$ is any function, then $f = f \circ \text{inc}_{A \rightarrow A} = \text{inc}_{B \rightarrow B} \circ f$.
- (c) [3 Marks] If $f : A \rightarrow B$ is bijective, then $f \circ f^{-1} = \text{inc}_{B \rightarrow B}$ and $f^{-1} \circ f = \text{inc}_{A \rightarrow A}$.
- (d) [3 Marks] If X and Y are disjoint sets and $f : X \rightarrow Z$, $g : Y \rightarrow Z$ are functions, then there is a unique function $h : X \cup Y \rightarrow Z$ such that $h \circ \text{inc}_{X \rightarrow X \cup Y} = f$ and $h \circ \text{inc}_{Y \rightarrow X \cup Y} = g$.

Solution. Recall that a function carries its domain and its range — the set Y in the notation $f : X \rightarrow Y$, which other texts call the codomain — as part of its data, and that two functions are equal precisely when they have the same domain, the same range, and the same value at every point of that common domain. Each part below therefore has two halves: check that the two sides are functions between the same pair of sets, then compare values. Note also that $\text{inc}_{X \rightarrow Y}$ requires $X \subseteq Y$ in order to exist at all, since the prescription $x \mapsto x$ must produce an element of Y for every $x \in X$.

(a) Both sides are functions from X to Z . For the left side, $\text{inc}_{X \rightarrow Y}$ has domain X and range Y , and $\text{inc}_{Y \rightarrow Z}$ has domain Y and range Z , so the composite is defined and goes from X to Z . For the right side, $X \subseteq Y$ and $Y \subseteq Z$ give $X \subseteq Z$, so $\text{inc}_{X \rightarrow Z}$ exists and also goes from X to Z . For $x \in X$,

$$(\text{inc}_{Y \rightarrow Z} \circ \text{inc}_{X \rightarrow Y})(x) = \text{inc}_{Y \rightarrow Z}(\text{inc}_{X \rightarrow Y}(x)) = \text{inc}_{Y \rightarrow Z}(x) = x = \text{inc}_{X \rightarrow Z}(x),$$

where the second equality uses $\text{inc}_{X \rightarrow Y}(x) = x$, and this is a legitimate input to $\text{inc}_{Y \rightarrow Z}$ because $x \in X \subseteq Y$. Same domain, same range, same values; hence the two are equal.

(b) All three are functions from A to B : $\text{inc}_{A \rightarrow A}$ has domain and range A , so $f \circ \text{inc}_{A \rightarrow A}$

goes from A to B , and $\text{inc}_{B \rightarrow B}$ has domain and range B , so $\text{inc}_{B \rightarrow B} \circ f$ also goes from A to B . For $x \in A$,

$$(f \circ \text{inc}_{A \rightarrow A})(x) = f(\text{inc}_{A \rightarrow A}(x)) = f(x),$$

and, since $f(x) \in B$,

$$(\text{inc}_{B \rightarrow B} \circ f)(x) = \text{inc}_{B \rightarrow B}(f(x)) = f(x).$$

So all three agree at every point of A , and all three go from A to B .

(c) Since f is bijective, $f^{-1} : B \rightarrow A$ exists, and for $y \in B$ the value $f^{-1}(y)$ is by definition the unique $x \in A$ with $f(x) = y$; surjectivity gives existence of such an x and injectivity gives its uniqueness. Hence $f \circ f^{-1}$ and $\text{inc}_{B \rightarrow B}$ are both functions from B to B , and $f^{-1} \circ f$ and $\text{inc}_{A \rightarrow A}$ are both functions from A to A . For every $y \in B$, taking $x := f^{-1}(y)$ so that $f(x) = y$,

$$(f \circ f^{-1})(y) = f(f^{-1}(y)) = f(x) = y = \text{inc}_{B \rightarrow B}(y),$$

and for every $x \in A$, the element $f^{-1}(f(x))$ is by definition the unique element sent by f to $f(x)$, which is x , so

$$(f^{-1} \circ f)(x) = x = \text{inc}_{A \rightarrow A}(x).$$

(d) *Existence.* Consider the property $P(w, z)$ given by

$$(w \in X \wedge z = f(w)) \vee (w \in Y \wedge z = g(w)).$$

We check that for each $w \in X \cup Y$ there is exactly one $z \in Z$ with $P(w, z)$. There is at least one, because w lies in X or in Y , and $f(w)$ respectively $g(w)$ is then an element of Z . There is at most one: if $w \in X$ only, the first disjunct alone can hold and it determines $z = f(w)$; similarly if $w \in Y$ only; and the case $w \in X \cap Y$, in which the two disjuncts might demand different values, does not arise, because X and Y are disjoint. So the property defines a function

$$h : X \cup Y \rightarrow Z, \quad h(w) = \begin{cases} f(w) & \text{if } w \in X, \\ g(w) & \text{if } w \in Y. \end{cases}$$

Disjointness is used at exactly this one point, to rule out a conflicting pair of assignments at a point of $X \cap Y$.

Now $X \subseteq X \cup Y$, so $\text{inc}_{X \rightarrow X \cup Y}$ exists, and both $h \circ \text{inc}_{X \rightarrow X \cup Y}$ and f are functions from X to Z . For $x \in X$,

$$(h \circ \text{inc}_{X \rightarrow X \cup Y})(x) = h(x) = f(x),$$

using the first case of the definition of h . The identity $h \circ \text{inc}_{Y \rightarrow X \cup Y} = g$ follows in the same way from the second case.

Uniqueness. Suppose h' also satisfies both identities. Let $w \in X \cup Y$. If $w \in X$, then

$$h'(w) = h'(\text{inc}_{X \rightarrow X \cup Y}(w)) = (h' \circ \text{inc}_{X \rightarrow X \cup Y})(w) = f(w) = h(w),$$

and if $w \in Y$ the same computation with g in place of f gives $h'(w) = g(w) = h(w)$. Since h and h' have the same domain $X \cup Y$ and the same range Z and agree at every point, $h = h'$. $\square$

Remark. The two halves of (d) do not stand or fall together. Existence used disjointness; uniqueness used no hypothesis at all, only that every point of $X \cup Y$ lies in X or in Y . Indeed uniqueness survives without disjointness, whereas existence fails: with $X = Y = \{0\}$, $f(0) := 0$ and $g(0) := 0++$, the property P above is satisfied by two different values at $w = 0$, so no function is determined.